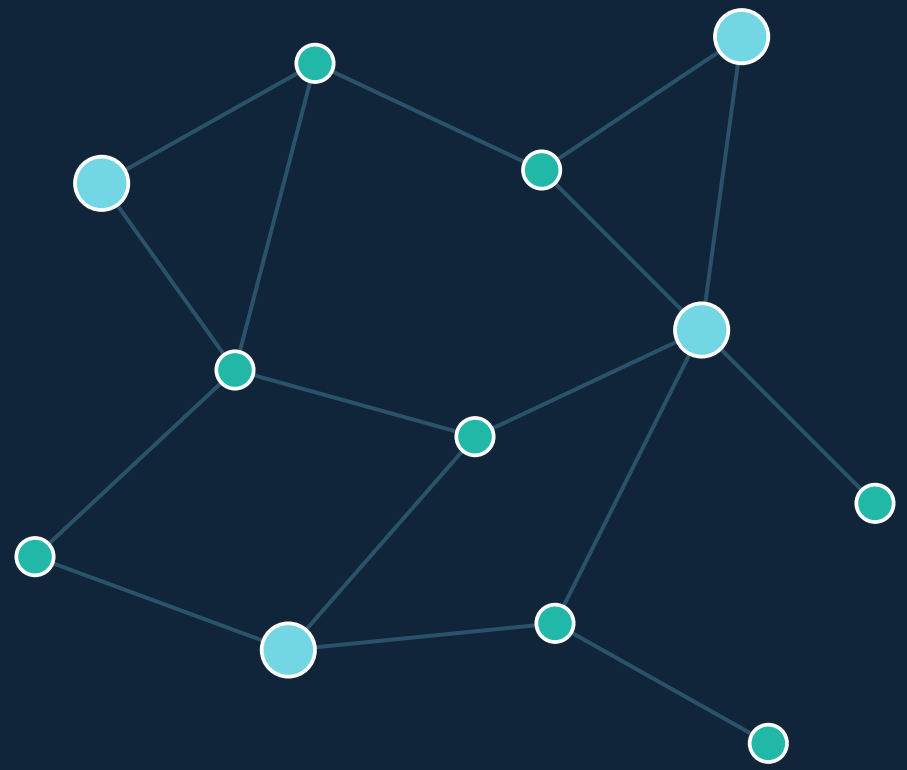
RESEARCH + EXPERIMENTAL FRAMEWORK

UrbanFlow AI

Intelligent Traffic Forecasting using Spatio-Temporal Graph Neural Networks



Dataset: METR-LA • 207 traffic sensors

5-minute sampling • 12-step history → 3-step forecast

Your Name | Guide Name | Department

Problem Statement

Why traffic forecasting needs both spatial and temporal reasoning

Traffic conditions depend on

- Temporal dependencies — previous traffic affects future traffic.
- Spatial dependencies — connected sensors influence one another.
- Temporal context — time of day and day of week affect traffic behaviour.
- Uncertainty — predictions may not always be equally reliable.

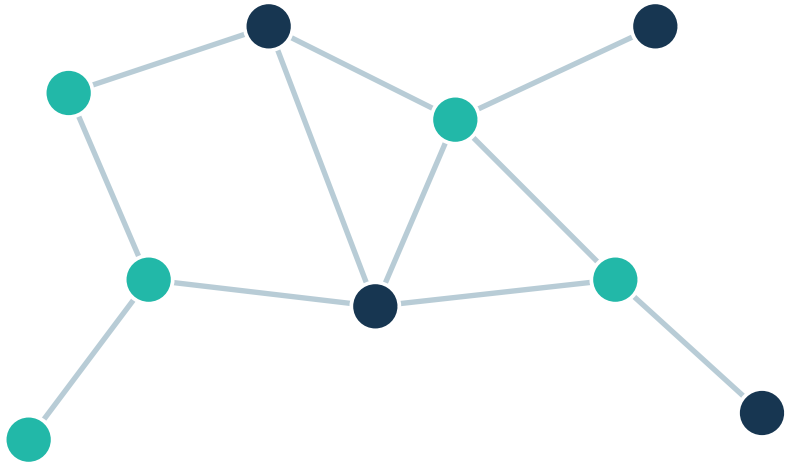
OBJECTIVE

Forecast future traffic conditions across interconnected sensors using Spatio-Temporal Graph Neural Networks.

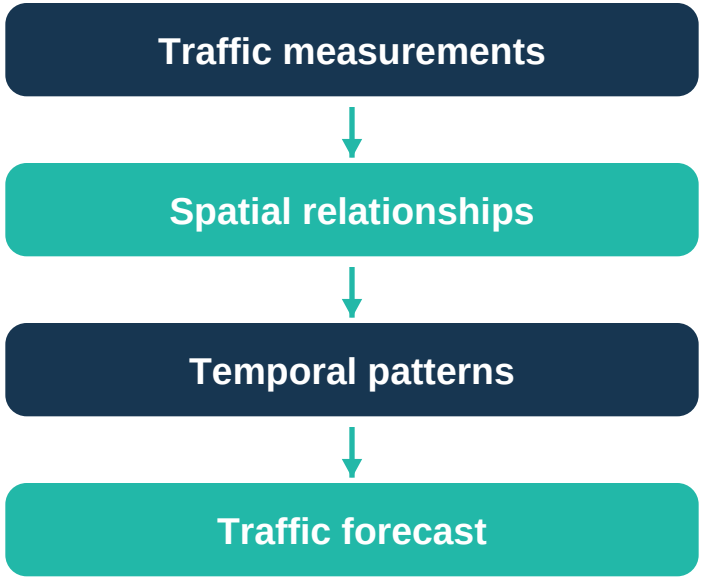
Why STGNN?

Traffic is naturally represented as a network that changes over time

Traffic sensors as a graph



Nodes = sensors Edges = sensor/road relationships



Literature Study

The review helped narrow broad research areas into testable questions

What existing research already covers

- **Dynamic / adaptive graphs**
- **Multi-scale temporal modelling**
- **Transformer architectures**
- **Long-term forecasting**
- **External/contextual features**
- **Robustness and missing data**
- **Scalability**
- **Frequency-domain modelling**

Key observation: these are active research areas, so the project tests specific questions rather than claiming broad novelty.

Research Questions

Three experimental questions guide the project

RQ1 • MULTI-SCALE

Can multiple temporal receptive fields improve traffic forecasting?

RQ2 • RELIABILITY

Can predictive uncertainty provide useful information about forecast reliability?

RQ3 • CONTEXT

Does temporal context improve forecasting beyond traffic history and graph structure?

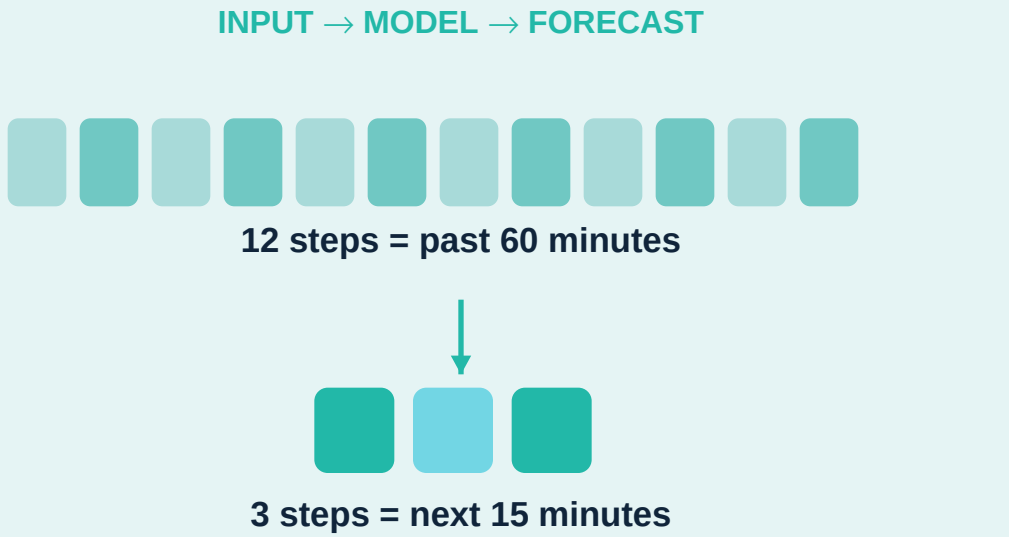
These questions are investigated experimentally on a consistent METR-LA pipeline.

Dataset

METR-LA traffic measurements and the fixed forecasting setup

Dataset properties

- 207 traffic sensors
- 5-minute sampling interval
- March–June 2012 coverage
- Chronological split: 70% train / 10% validation / 20% test



Complete Experimental Pipeline

The notebooks build the experiment from data validation to final analysis

- 01 Dataset exploration
- 02 Preprocessing
- 03 Baselines
- 04 Standard STGCN
- 05 Evaluation
- 06 Multi-Scale STGCN
- 07 Error analysis
- 08 Uncertainty estimation
- 09 Abnormal-condition testing
- 10 Context-aware forecasting

Each notebook answers a specific part of the experimental workflow.

What We Found

Summary of the evidence from the three research questions

RQ1 — Multi-scale temporal modelling

Compared kernels (3), (5), and (3,5). • Different temporal receptive fields affected forecasting performance. • The experiments do not establish universal superiority.

RQ2 — Reliability

MC-Dropout uncertainty had a moderate relationship with forecasting error. • Raw prediction intervals were poorly calibrated. • Validation-based calibration brought test coverage close to the intended 95%.

RQ3 — Context

Compared traffic-only, direct concatenation, and context-gated models. • Gating reduced RMSE relative to the traffic-only baseline. • Improvements were not consistent across MAE, RMSE, and MAPE.

What Is Still Not Solved

Limitations matter as much as positive results

Remaining research gaps

- Multi-scale modelling was investigated, but not shown to be universally superior.
- Uncertainty estimates required calibration to obtain useful interval coverage.
- Context did not improve all evaluation metrics consistently.
- Stress testing showed progressive degradation under stronger perturbations.
- Experiments were conducted on METR-LA with controlled synthetic perturbations.

Next validation steps

- Test on additional datasets.
- Evaluate on real abnormal traffic events.
- Study calibration across settings.
- Investigate context integration further.

Conclusion

What UrbanFlow AI has achieved

EXPERIMENTAL PIPELINE

- Validated and preprocessed METR-LA
- Established classical and deep-learning baselines
- Implemented standard and multi-scale STGCN

RELIABILITY + BEHAVIOUR

- Analysed errors by horizon and sensor
- Estimated uncertainty and calibrated intervals
- Stress-tested controlled abnormal inputs

CONTEXT + EVIDENCE

- Compared concatenation and gating
- Reported mixed metric outcomes honestly
- Documented limitations and next steps

A reproducible framework for investigating traffic forecasting — not a claim that every research problem is solved.

Future Work & Closing

A focused path from current experiments to stronger evidence

Future work

- Validate the findings on additional traffic datasets.
- Use real incident or abnormal-condition data where available.
- Improve uncertainty calibration and test its transferability.
- Explore more effective integration of temporal context.
- Evaluate performance and computational cost under broader settings.

FINAL TAKEAWAY

UrbanFlow AI combines a reproducible forecasting pipeline with controlled experiments on multi-scale modelling, uncertainty, stress sensitivity, and context.

Thank you.